

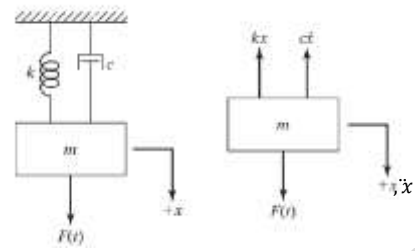
Harmonically Excited Vibration of viscous damped system

Consider a single degree of freedom spring-mass-dashpot system subjected to a harmonic force

$$F(t) = F_0 \sin \omega t$$

From the freebody diagram, we can write the equation of motion as

$$m\ddot{x} = F_0 \sin \omega t - c\dot{x} - kx \quad \Leftrightarrow \quad m\ddot{x} + c\dot{x} + kx = F_0 \sin \omega t$$



The governing equation is a second order, non homogeneous, linear, ordinary differential equation with constant coefficients. The solution will have two parts; a homogeneous solution and a particular solution. The homogeneous solution is obtained from the homogeneous part $m\ddot{x} + c\dot{x} + kx = 0$

Homogeneous solution:

$$x_h(t) = (A_1 + A_2 t)e^{-\omega_n t}, \quad \text{if } \zeta = 1$$

$$x_h(t) = e^{-\zeta \omega_n t} (A_1 e^{\omega_n \sqrt{\zeta^2 - 1} t} + A_2 e^{-\omega_n \sqrt{\zeta^2 - 1} t}), \quad \text{if } \zeta > 1$$

$$x_h(t) = X_0 e^{-\zeta \omega_n t} \sin(\omega_d t + \varphi), \quad \text{if } \zeta < 1$$

Particular solution:

Since the excitation is harmonic, let us assume the particular response also as harmonic.

$$x_p(t) = C_1 \sin \omega t + C_2 \cos \omega t \quad \Leftrightarrow \quad \dot{x}_p = \omega(C_1 \cos \omega t - C_2 \sin \omega t) \quad \Leftrightarrow \quad \ddot{x}_p = -\omega^2(C_1 \sin \omega t + C_2 \cos \omega t)$$

Substituting this solution back in the differential equation, we get

$$m\{-\omega^2(C_1 \sin \omega t + C_2 \cos \omega t)\} + c\{\omega(C_1 \cos \omega t - C_2 \sin \omega t)\} + k\{C_1 \sin \omega t + C_2 \cos \omega t\} = F_0 \sin \omega t$$

$$\{(k - m\omega^2)C_1 - c\omega C_2\} \sin \omega t + \{(k - m\omega^2)C_2 + c\omega C_1\} \cos \omega t = F_0 \sin \omega t$$

Equating the coefficients of $\sin \omega t$ and $\cos \omega t$ on both sides, we get

$$\begin{aligned} (k - m\omega^2)C_1 - c\omega C_2 &= F_0 \\ c\omega C_1 + (k - m\omega^2)C_2 &= 0 \end{aligned}$$

$$\begin{bmatrix} k - m\omega^2 & -c\omega \\ c\omega & k - m\omega^2 \end{bmatrix} \begin{Bmatrix} C_1 \\ C_2 \end{Bmatrix} = \begin{Bmatrix} F_0 \\ 0 \end{Bmatrix}$$

Solving these two equations, the two constants can be evaluated.

$$C_1 = \frac{\begin{vmatrix} F_0 & -c\omega \\ 0 & k - m\omega^2 \end{vmatrix}}{\begin{vmatrix} k - m\omega^2 & -c\omega \\ c\omega & k - m\omega^2 \end{vmatrix}} = \frac{F_0(k - m\omega^2)}{(k - m\omega^2)^2 + (c\omega)^2}$$

$$C_2 = \frac{\begin{vmatrix} k - m\omega^2 & F_0 \\ c\omega & 0 \end{vmatrix}}{\begin{vmatrix} k - m\omega^2 & -c\omega \\ c\omega & k - m\omega^2 \end{vmatrix}} = \frac{F_0(-c\omega)}{(k - m\omega^2)^2 + (c\omega)^2}$$

$$x_p(t) = C_1 \sin \omega t + C_2 \cos \omega t$$

$$x_p(t) = \frac{F_0(k - m\omega^2)}{(k - m\omega^2)^2 + (c\omega)^2} \sin \omega t + \frac{F_0(-c\omega)}{(k - m\omega^2)^2 + (c\omega)^2} \cos \omega t$$

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$$x_p(t) = \frac{F_0}{(k - m\omega^2)^2 + (c\omega)^2} \{(k - m\omega^2) \sin \omega t - c\omega \cos \omega t\}$$

$$k - m\omega^2 = X_0 \cos \theta \quad \Leftrightarrow \quad c\omega = X_0 \sin \theta$$

$$X_0 = \sqrt{(k - m\omega^2)^2 + (c\omega)^2} = k \sqrt{\left(1 - \frac{\omega^2}{k/m}\right)^2 + \left(\frac{c\omega}{k}\right)^2} = k \sqrt{(1 - r^2)^2 + (2r\zeta)^2}$$

$$\tan \theta = \frac{c\omega}{k - m\omega^2} \quad \Leftrightarrow \quad \tan \theta = \frac{\frac{c\omega}{k}}{1 - r^2} = \frac{2r\zeta}{1 - r^2}$$

$$x_p(t) = \frac{F_0/k^2}{(1 - r^2)^2 + (2r\zeta)^2} \{X_0 \cos \theta \sin \omega t - X_0 \sin \theta \cos \omega t\} = \frac{F_0/k^2}{(1 - r^2)^2 + (2r\zeta)^2} X_0 \sin(\omega t - \theta)$$

$$x_p(t) = \frac{F_0/k^2}{(1 - r^2)^2 + (2r\zeta)^2} k \sqrt{(1 - r^2)^2 + (2r\zeta)^2} \sin(\omega t - \theta) = \frac{F_0/k}{\sqrt{(1 - r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \theta)$$

The complete solution is

$$x(t) = (A_1 + A_2 t)e^{-\omega_n t} + \frac{F_0/k}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \theta), \quad \text{critically damped}$$

$$x(t) = e^{-\zeta \omega_n t} (A_1 e^{\omega_n \sqrt{\zeta^2 - 1} t} + A_2 e^{-\omega_n \sqrt{\zeta^2 - 1} t}) + \frac{F_0/k}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \theta), \quad \text{overdamped}$$

$$x(t) = X_0 e^{-\zeta \omega_n t} \sin(\omega_d t + \phi) + \frac{F_0/k}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \theta), \quad \text{underdamped}$$

We can see that the homogeneous solution tends to zero as time elapses and the steady state solution

$$x_{ss} = \frac{F_0/k}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \theta),$$

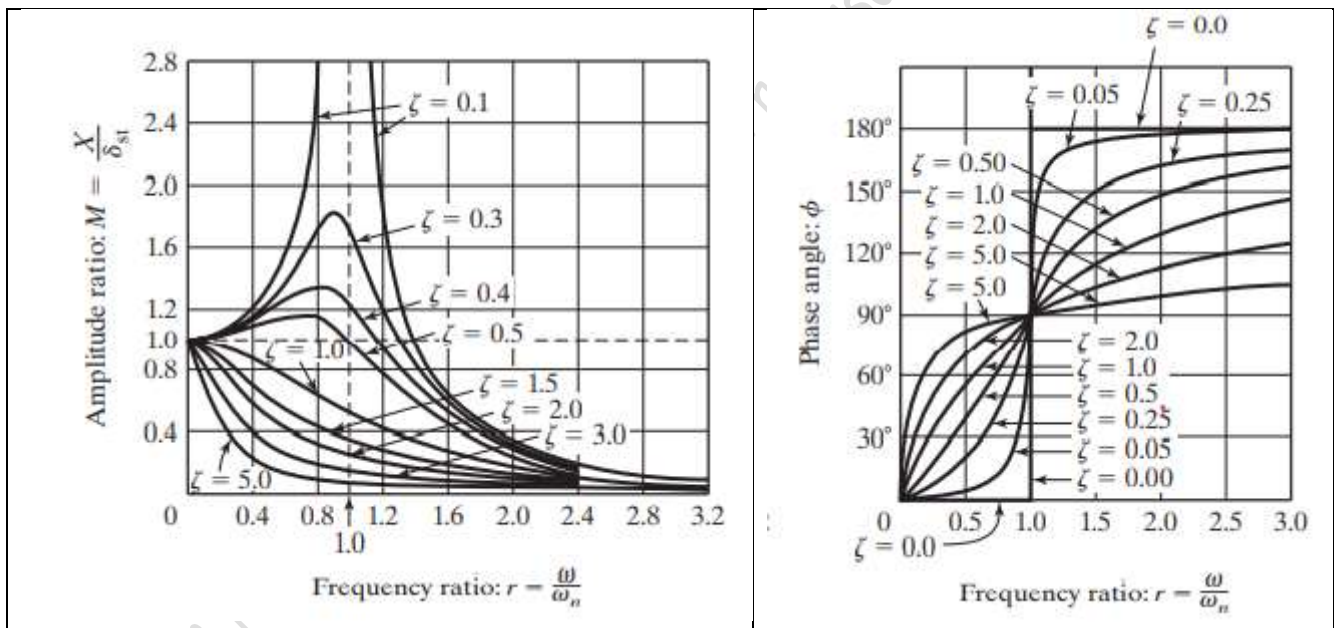
The amplitude of steady state vibration is

$$X = \frac{F_0/k}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}}$$

The magnification factor,

$$\beta = \frac{X}{F_0/k} = \frac{1}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}}$$

The magnification factor β and phase angle ϕ are plotted against the frequency ratio r .



The maximum value of β is obtained by considering $\frac{d\beta}{dr} = 0$.

$$\frac{1}{2\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \times \{2(1-r^2)(-2r) + 4\zeta^2(2r)\} = 0 \quad \Leftrightarrow \quad 2(1-r^2) = 4\zeta^2 \quad \Leftrightarrow \quad r = \sqrt{1-2\zeta^2}$$

This is valid only for $0 < \zeta < \frac{1}{\sqrt{2}}$.

$$\text{The maximum value of } \beta = \left| \frac{1}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \right|_{r=\sqrt{1-2\zeta^2}} = \frac{1}{\sqrt{(2\zeta^2)^2 + 4\zeta^2(1-2\zeta^2)}} = \frac{1}{\sqrt{4\zeta^4 + 4\zeta^2 - 8\zeta^4}} = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

This equation can be used to find the value of damping factor in a system experimentally.

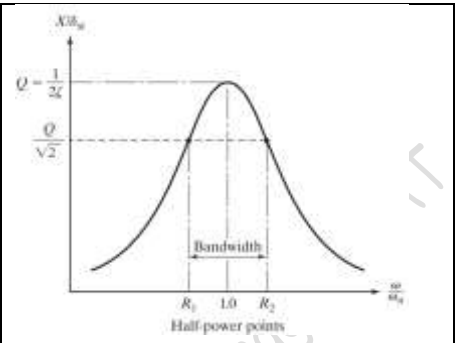
When $r = 1$, the value of $\beta = \frac{1}{2\zeta}$. The value of magnitude ratio at resonance is known as Quality factor or Q-factor of the system.

$$Q = \frac{1}{2\zeta}$$

The points R_1 and R_2 where the amplification factor falls to $\frac{Q}{\sqrt{2}}$ is called half power points, because the power (ΔW) absorbed by the damper (or by the resistor in an electrical circuit), responding harmonically, is proportional to the square of the amplitude.

$$\Delta W = \pi c \omega X^2$$

The difference between frequencies associated with the half power points R_1 and R_2 is called the bandwidth of the system. The reduction in gain will be **3 dB**



To find R_1 and R_2 , we can substitute $\beta = \frac{Q}{\sqrt{2}}$

$$\frac{1}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} = \frac{Q}{\sqrt{2}} = \frac{1}{\sqrt{2}} \frac{1}{2\zeta} \quad \Leftrightarrow \quad r^4 - r^2(2 - 4\zeta^2) + (1 - 8\zeta^2) = 0$$

$$r_1^2 = 1 - 2\zeta^2 - 2\zeta\sqrt{1 + \zeta^2} \quad \Leftrightarrow \quad r_2^2 = 1 - 2\zeta^2 + 2\zeta\sqrt{1 + \zeta^2}$$

For small values of ζ , $r_1^2 \cong 1 - 2\zeta$ and $r_2^2 \cong 1 + 2\zeta$

$$r_1^2 = R_1^2 = \left(\frac{\omega_1}{\omega_n}\right)^2 = 1 - 2\zeta \quad \Leftrightarrow \quad r_2^2 = R_2^2 = \left(\frac{\omega_2}{\omega_n}\right)^2 = 1 + 2\zeta$$

$$R_2^2 - R_1^2 = 4\zeta \quad \Leftrightarrow \quad \left(\frac{\omega_2}{\omega_n}\right)^2 - \left(\frac{\omega_1}{\omega_n}\right)^2 = 4\zeta \quad \Leftrightarrow \quad \omega_2^2 - \omega_1^2 = 4\zeta\omega_n^2$$

$$(\omega_2 - \omega_1) \frac{(\omega_2 + \omega_1)}{2} = 2\zeta\omega_n^2 \quad \Leftrightarrow \quad (\omega_2 - \omega_1)\omega_n = 2\zeta\omega_n^2 \quad \Leftrightarrow \quad (\omega_2 - \omega_1) = 2\zeta\omega_n$$

$$Q = \frac{1}{2\zeta} = \frac{\omega_n}{(\omega_2 - \omega_1)}$$

Force transmitted to the base

The force transmitted to the base, $f_T = c\dot{x} + kx$

Assuming steady state, $x(t) = \frac{\frac{F_0}{k}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \theta) \quad \Leftrightarrow \quad \dot{x}(t) = \frac{\frac{F_0}{k}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \omega \cos(\omega t - \theta)$

$$f_T = c\omega \frac{\frac{F_0}{k}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \cos(\omega t - \theta) + k \frac{\frac{F_0}{k}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \theta)$$

$$f_T = \frac{\frac{F_0}{k}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \{c\omega \cos(\omega t - \theta) + k \sin(\omega t - \theta)\}$$

$$f_T = \frac{\frac{F_0}{k}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} A \{\sin \psi \cos(\omega t - \theta) + \cos \psi \sin(\omega t - \theta)\}$$

$$c\omega = A \sin \psi \quad \Leftrightarrow \quad k = A \cos \psi$$

$$A = \sqrt{k^2 + (c\omega)^2} \quad \Leftrightarrow \quad A = k \sqrt{1 + \left(\frac{c\omega}{k}\right)^2}$$

$$A = k \sqrt{1 + (2r\zeta)^2} \quad \Leftrightarrow \quad \tan \psi = \frac{c\omega}{k} = 2r\zeta$$

$$f_T = \frac{\frac{F_0}{k} k \sqrt{1 + (2r\zeta)^2}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \theta + \psi) = \frac{F_0 \sqrt{1 + (2r\zeta)^2}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \theta + \psi) = \frac{F_0 \sqrt{1 + (2r\zeta)^2}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \alpha)$$

The amplitude of force transmitted to the base at steady state,

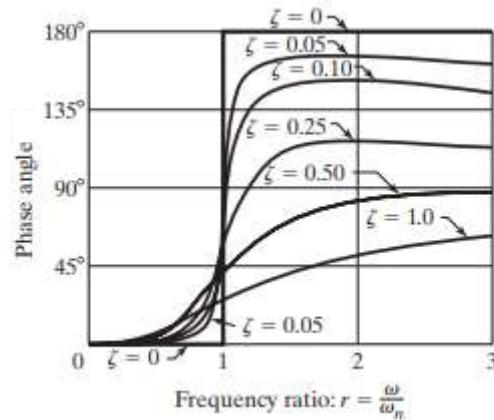
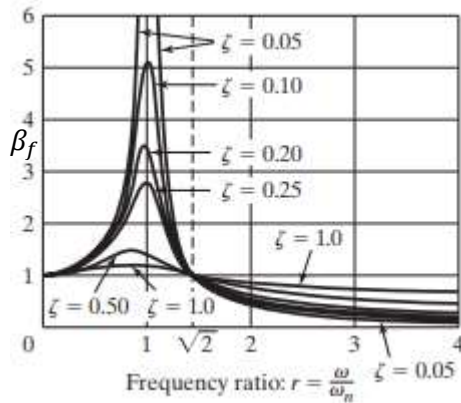
$$F_T = \frac{F_0 \sqrt{1 + (2r\zeta)^2}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}}$$

The ratio of amplitude of dynamic force transmitted to the base to the amplitude of excitation force is known as **force transmissibility** β_f

$$\beta_f = \frac{F_T}{F_0} = \frac{\sqrt{1 + (2r\zeta)^2}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}}$$

The phase angle $\alpha = \theta - \psi \quad \ll \gg \quad \tan \alpha = \frac{\tan \theta - \tan \psi}{1 + \tan \theta \tan \psi} = \frac{\frac{2r\zeta}{1-r^2} - 2r\zeta}{1 + \frac{2r\zeta}{1-r^2} 2r\zeta} = \frac{2\zeta r^3}{1-r^2+4\zeta^2 r^2}$

The force transmissibility β_f and phase angle α are plotted against frequency ratio r as shown.



#Example 1

Find the total response of a single degree of freedom system with $m = 10 \text{ kg}$, $c = 20 \text{ N}\cdot\frac{s}{m}$, $k = 4000 \frac{N}{m}$, $x_0 = 0.01 \text{ m}$, and $\dot{x}_0 = 0$ when a force of $100 \sin 10t \text{ N}$ is acting on the system. What is the amplification factor and force transmissibility?

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{4000}{10}} = 20 \text{ rad/s}$$

$$c = 2m\zeta\omega_n \ll \gg \zeta = \frac{20}{2 \times 10 \times 20} = 0.05$$

The system is under damped.

$$F_0 = 100 \text{ N}, \quad \omega = 10 \frac{\text{rad}}{\text{s}} \quad \ll \gg \quad r = \frac{\omega}{\omega_n} = 0.5$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = 19.98 \text{ rad/s}$$

$$x(t) = X_0 e^{-\zeta\omega_n t} \sin(\omega_d t + \varphi) + \frac{F_0/k}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \theta),$$

$$\tan \theta = \frac{2r\zeta}{1-r^2} \ll \gg \theta = 3.8 \text{ rad} = 217.7^\circ$$

$$x(t) = X_0 e^{-0.5 \times 20 t} \sin(19.98 t + \varphi) + \frac{100/4000}{\sqrt{(1-0.5^2)^2 + (2 \times 0.5 \times 0.05)^2}} \sin(10 t - 3.8),$$

$$x(t) = X_0 e^{-10t} \sin(19.98 t + \varphi) + 0.03326 \sin(10 t - 3.8)$$

$$\dot{x} = X_0 e^{-10t} \times 19.98 \cos(19.98 t + \varphi) + X_0 \sin(19.98 t + \varphi)(-10) e^{-10t} + 0.3326 \cos(10 t - 3.8)$$

When $t = 0$, $\dot{x} = 0$, and $x = 0.01$

$$0 = 19.98 X_0 \cos(\varphi) - 10 X_0 \sin(\varphi) - 0.8 \ll \gg 19.98 X_0 \cos(\varphi) - 10 X_0 \sin(\varphi) = 0.8$$

$$0.01 = X_0 \sin(\varphi) - 0.611 \ll \gg X_0 \sin(\varphi) = 0.621$$

$$X_0 \cos(\varphi) = 0.04 + 0.314 = 0.354 \gg \gg X_0 = \sqrt{0.621^2 + 0.354^2} = 0.714 \gg \gg \varphi = 60.3^\circ = 1.05 \text{ rad}$$

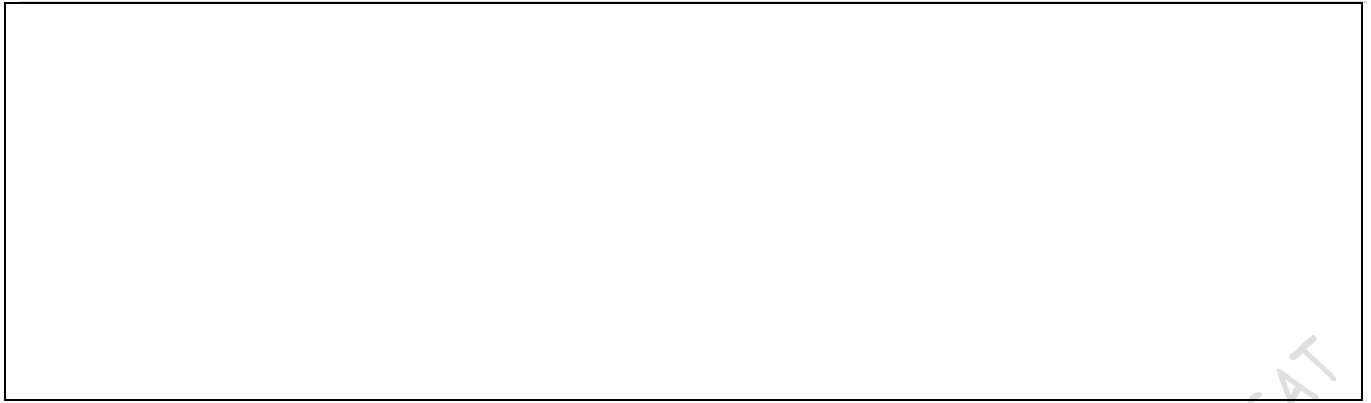
$$\text{Total response, } x(t) = 0.714 e^{-10t} \sin(19.98 t + 1.05) + 0.03326 \sin(10 t - 3.8)$$

$$\text{The amplification factor, } \beta = \frac{1}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} = 1.33$$

$$\text{The force transmissibility, } \beta_f = \frac{\sqrt{1+(2r\zeta)^2}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} = 1.33$$

Reference:

Mechanical Vibrations, 6/e, Singiresu S. Rao, Pearson Global Edition



Vibration & Noise Control..... Dr. Biju N., Professor, School of Engg, CUSAT